Extended piston rods have been tried out with more or less success on a large number of American roads as a means of reducing cylinder and piston wear. Any means employed for relieving the cylinder and piston from the weight, shock and consequent wear attendant upon a free piston, would seem to be a direct benefit. If it were always possible to preserve the alignment on both ends of the rod the extended piston rod would be beneficial in preventing cylinder wear and would also permit the use of cast steel, thereby resulting in some reduction in weight. On the other hand the extended rod itself results in considerable increase in weight, which seriously complicates the question of counter balancing.

Two general methods have been resorted to in treating the extended piston rod; one, the use of another metallic packing on the front cylinder head, while the other employs an extension sheath in which the piston rod is permitted to reciprocate the bronze or other suitable form of bearing at the inner end of this sheath to take the weight of the rod and piston. In the former case the upkeep of an extra metallic packing is a serious objection, while in the latter case the steam clearance is considerably increased and it is difficult to design a bushing of sufficient bearing to prevent rapid wear. The committee are convinced that there are diverse opinions as to the merits of extended piston ends, hence definite conclusions should be deferred and more investigation made on the subject.

Regarding solid bushed front end main rods, the builders state that several years ago quite a number of roads had locomotives equipped with solid front end main rod and bushing, but have since gone back to the use of keys and two-piece brass on account of the solid bushing not giving entirely satisfactory service.

With reference to the comparative merits of the Mikado and Consolidation types of locomotives, the first Mikado types were built by the Baldwin Locomotive Works in 1897 for the Nippon Railway of Japan; in 1902 a locomotive of similar type was built for the Bismarck, Washburn and Great Falls Railway in the United States.

The Mikado type locomotive permits the use of a long boiler with a wide and deep firebox, which is placed back of the driving wheels and over the trailing truck. This form of firebox is especially suitable for burning high volatile coal, as there is sufficient furnace volume for the combustion of the gases before they enter the tubes. Cases are on record where Mikado type locomotives have successfully burned fuel of a quality too poor for use in a Consolidation type engine of equivalent hauling capacity, but having smaller fireboxes. With a given weight on driving wheels and equal ratios of adhesion, a Mikado type locomotive will show no superiority over a Consolidation type as far as starting effort is concerned. As the speed increases, however, the tractive effort of the Mikado type will fall off less rapidly than that of the Consolidation, because of the greater boiler power of the former locomotive. On some roads the Mikado type locomotive has made an excellent combined passenger and freight locomotive, being used interchangeably for either service as occasion requires. The Mikado engine, on account of the wheel base arrangement, is somewhat easier on flange wear, and somewhat safer in high speed without danger of derailment. On Consolidation type locomotives having wide fireboxes, and comparatively large driving wheels, the depth of the furnace throat is necessarily restricted, making it difficult to apply a satisfactory design of brick arch. This difficulty is avoided in the Mikado type, as there is ample space between the grate and bottom row of flues for an arch with its supporting tubes. The necessity of placing a driving wheel under the firebox makes it practically impossible to apply a satisfactory ash pan to the Consolidation type, because the pocket over the driving wheel is so close to the underside of the grates.

The committee further reported that they had issued a questionnaire to sixteen railroads in regard to the relative advantages of non-lifting and lifting injectors. The advantages of the non-lifting injectors are reported as follows: Larger capacity for same size injector, injector pipes and connections; grades closer, giving a wider range for each size of injector; gives less trouble from restricted openings in the feed water supply pipe; located on the outside of the cab, thus relieving congested conditions inside; more reliable and dependable; application and maintenance cost less; more accessible for repairs and repairs can be made on the injector in place without disconnecting from the piping; will work with less water in tank, also operate with feed water at a considerably higher temperature.

The advantages of this type of injector are its greater liability to being knocked off by derailment, side swipe, etc., and the liability to freeze in extremely cold weather.

The advantages of the lifting injectors are stated as follows:

Mounted in cab where engineman can more easily see and observe its workings; if trouble occurs on the road it is easier for the engineer to remedy it; self contained and fewer parts to operate; easier operated and controlled.

Its disadvantages are: Less grading range than the non-lifter; more readily affected by restricted feed water opening and length of lift; must be taken down from pipe connections to repair the tubes; tubes coat up more rapidly than in the non-lifters; it is subject to failure from a leaky boiler check, worn steam nozzle or combining tube, leak in feed pipe, or if injectors become overheated they must be cooled off before they will operate.

It would appear from the replies received that: Little is known about the comparative efficiency of the two types of injectors with respect to steam consumption for a given quantity of water delivered; the cost of repairs and maintenance is less with the non-lifting injector; the non-lifting injector is more easily located and repaired; the non-lifting injector is generally considered preferable and is gaining in favor, particularly on large locomotives.

For the purpose of securing data on the subject of saturated steam drifting devices for superheater locomotives a questionnaire was prepared and sent to 16 railroads from which 14 replies have been received. It would appear from replies received that:

Automatic drifting valves on superheater locomotives are not generally being used; the common practice when drifting is to crack the throttle; the use of the manually controlled steam line from cab to valves and cylinders has not proven satisfactory; some roads are using home made devices for supplying air compressor exhaust steam to valves and cylinders with satisfactory results; a few roads are using automatic drifting valves with apparent success; a few roads are experimenting with various forms of automatic drifting valves; an automatic drifting valve is desirable if one can be found or developed which will operate successfully both at high and at low speeds, and be simple and rugged enough to be reliable and stand the service requirements without undue maintenance cost.

In order to determine modern practice of representative railroads in regard to tank syphons versus tank valves, a questionnaire was prepared and sent to 24 railroads, from which 16 replies have been received. Summarizing the data submitted by the various railroads, it appeared to the committee that the syphon is well adapted for use with lifting injectors, while the tank valve is more suitable for non-lifting injectors.

To the inquiry regarding the subject of tank hose versus metallic connections no replies were received from railroads and one voluntary report from a manufacturer. All of these roads but one, were large systems. Five roads had no experience whatever with metallic tank connections. One road reported trying out one in 1909 without success. Another road has tried one on a switch engine for three years with no cost for repairs beyond three or four gaskets and their application. The Erie has one connection applied to engine No. 2509. This has been in service for the last 18 months. The test is to cover a period of two years. One large system reported having had "a little experience" with metallic tank connections, adding "We are not extending it."

From our study of this subject it ap-

remaining; on others there has been no diminution of pressure, and the last condition of those cars is worse than the first. This means, cycling, recharging, shocks and the making of excessive stresses between cars with possible break-in-twos.

The report deals in detail with the investigations at the several terminals with the following general results, showing the number of cars falling within the limits of the piston travel and leakage per minute from 50 lbs. as specified.

TABLE No. 1

Piston travel 3½- 4 in.	3
Piston travel 4¼- 5 in.	9
Piston travel 5¼- 6 in.	35
Piston travel 6¼- 7 in.	28
Piston travel 7¼- 8 in.	11
Piston travel 8¼- 9 in.	6
Piston travel 9¼-10 in.	4
Piston travel 10¼-11 in.	3
Piston travel 11¼-12 in.	1
Leakage 5 lbs. or less	24
Leakage 5.5-10 lbs.	16
Leakage 10.5-15 lbs.	11
Leakage 16-20 lbs.	16
Leakage 21-25 lbs.	6
Leakage 26-30 lbs.	13
Leakage 31-35 lbs.	3
Leakage 36-40 lbs.	3
Leakage 41-45 lbs.	1
Leakage 46-50 lbs.	7

TABLE No. 2

Piston travel 3½- 4 in.	2
Piston travel 4¼- 5 in.	17
Piston travel 5¼- 6 in.	21
Piston travel 6¼- 7 in.	27
Piston travel 7¼- 8 in.	17
Piston travel 8¼- 9 in.	12
Piston travel 9¼-10 in.	4
Leakage 5 lbs. or less	21
Leakage 5.5-10 lbs.	19
Leakage 10.5-15 lbs.	15
Leakage 16-20 lbs.	11
Leakage 21-25 lbs.	14
Leakage 26-30 lbs.	4
Leakage 31-35 lbs.	4
Leakage 36-40 lbs.	3
Leakage 41-45 lbs.	0
Leakage 46-50 lbs.	9

TABLE No. 3

Piston travel 4¼- 5 in.	7
Piston travel 5¼- 6 in.	23
Piston travel 6¼- 7 in.	19
Piston travel 7¼- 8 in.	21
Piston travel 8¼- 9 in.	18
Piston travel 9¼-10 in.	8
Piston travel 10¼-11 in.	0
Piston travel 11¼-12 in.	4
Leakage 5 lbs. or less	41
Leakage 5.5-10 lbs.	6
Leakage 10.5-15 lbs.	14
Leakage 16-20 lbs.	10
Leakage 21-25 lbs.	11
Leakage 26-30 lbs.	4
Leakage 31-35 lbs.	4
Leakage 36-40 lbs.	3
Leakage 41-45 lbs.	3
Leakage 46-50 lbs.	4

TABLE No. 4

Piston travel 3½- 4 in.	0
Piston travel 4¼- 5 in.	8
Piston travel 5¼- 6 in.	17
Piston travel 6¼- 7 in.	23
Piston travel 7¼- 8 in.	23
Piston travel 8¼- 9 in.	13
Piston travel 9¼-10 in.	6
Piston travel 10¼-11 in.	7
Piston travel 11¼-12 in.	3
	100
Leakage 5 lbs. or less	21
Leakage 5.5-10 lbs.	15
Leakage 10.5-15 lbs.	18
Leakage 16-20 lbs.	11
Leakage 21-25 lbs.	12
Leakage 26-30 lbs.	5
Leakage 31-35 lbs.	6
Leakage 36-40 lbs.	4
Leakage 41-45 lbs.	2
Leakage 46-50 lbs.	6

TABLE No. 5

Piston travel 3½- 4 in.	2
Piston travel 4¼- 5 in.	9
Piston travel 5¼- 6 in.	17
Piston travel 6¼- 7 in.	28
Piston travel 7¼- 8 in.	18
Piston travel 8¼- 9 in.	14
Piston travel 9¼-10 in.	8
Piston travel 10¼-11 in.	3
Piston travel 11¼-12 in.	1
Leakage 5 lbs. or less	25
Leakage 5.5-10 lbs.	10
Leakage 10.5-15 lbs.	16
Leakage 16-20 lbs.	13
Leakage 21-25 lbs.	11
Leakage 26-30 lbs.	11
Leakage 31-35 lbs.	6
Leakage 36-40 lbs.	3
Leakage 41-45 lbs.	1
Leakage 46-50 lbs.	4

TABLE No. 6

Piston travel 3½- 4 in.	2
Piston travel 4¼- 5 in.	12
Piston travel 5¼- 6 in.	18
Piston travel 6¼- 7 in.	14
Piston travel 7¼- 8 in.	20
Piston travel 8¼- 9 in.	8
Piston travel 9¼-10 in.	8
Piston travel 10¼-11 in.	8
Piston travel 11¼-12 in.	10
Leakage 5 lbs. or less	19
Leakage 5.5-10 lbs.	22
Leakage 10.5-15 lbs.	10
Leakage 16-20 lbs.	8
Leakage 21-25 lbs.	8
Leakage 26-30 lbs.	7
Leakage 31-35 lbs.	6
Leakage 36-40 lbs.	3
Leakage 41-45 lbs.	2
Leakage 46-50 lbs.	15

TABLE No. 7

Piston travel 3½- 4 in.	1
Piston travel 4¼- 5 in.	8
Piston travel 5¼- 6 in.	22
Piston travel 6¼- 7 in.	10
Piston travel 7¼- 8 in.	19
Piston travel 8¼- 9 in.	5
Piston travel 9¼-10 in.	3
Piston travel 10¼-11 in.	2
	100
Leakage 5 lbs. or less	23
Leakage 5.5-10 lbs.	7
Leakage 10.5-15 lbs.	21
Leakage 16-20 lbs.	14
Leakage 21-25 lbs.	12
Leakage 26-30 lbs.	13
Leakage 31-35 lbs.	4
Leakage 36-40 lbs.	4
Leakage 41-45 lbs.	1
Leakage 46-50 lbs.	1

SUMMARY OF ALL LEAKAGE TABLES

Leakage 5 lbs. or less	174—24.9%
Leakage 5.5-10 lb.	95—13.5%
Leakage 10.5-15 lb.	105—15.0%
Leakage 16.0-20 lb.	83—11.9%
Leakage 21.0-25 lb.	74—10.6%
Leakage 26.0-30 lb.	57— 8.1%
Leakage 31.0-35 lb.	33— 4.7%
Leakage 36.0-40 lb.	23— 3.3%
Leakage 41.0-45 lb.	10— 1.4%
Leakage 46.0-50 lb.	46— 6.6%
	700—100.0%

AVERAGES OF LEAKING AND PISTON TRAVEL

	Piston Travel	Leakage (Per 50#)	Pressure Initial	Pressure After 1 Min.
Table 1.	6.62	17.32	25.69	16.26
Table 2.	6.61	17.25	25.85	15.19
Table 3.	7.20	14.33	22.80	14.73
Table 4.	7.45	17.55	22.57	13.14
Table 5.	7.19	17.02	23.52	13.68
Table 6.	7.71	20.08	21.78	10.97
Table 7.	6.75	16.53	24.91	14.38
Average	7.09	17.15	23.87	14.05

AVERAGE LEAKAGE FOR PRESSURE DEVELOPED

Table 1— 9.43#	Table 5— 9.84#
Table 2—10.66#	Table 6—10.81#
Table 3— 8.44#	Table 7—10.00#
Table 4— 9.43#	

AVERAGE LEAKAGES FOR PRESSURE DEVELOPED

Table 1	9.43
Table 2	10.66
Table 3	8.44
Table 4	9.43
Table 5	9.84
Table 6	10.81
Table 7	10.00

SUMMARY OF PISTON TRAVEL

Piston travel 3½- 4 in.	17	2.4%
Piston travel 4¼- 5 in.	86	12.3%
Piston travel 5¼- 6 in.	149	21.3%
Piston travel 6¼- 7 in.	181	25.9%
Piston travel 7¼- 8 in.	126	18.0%
Piston travel 8¼- 9 in.	66	9.4%
Piston travel 9¼-10 in.	33	4.7%
Piston travel 10¼-11 in	27	3.9%
Piston travel 11¼-12 in	15	2.1%
	700	100.0%

Diagrams are presented showing the variations in the leakage of the cars taken at one of the terminals represented at No. 4, from which it will be seen that the ef-